POLICY ARTICLE

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HIGHER EDUCATION

Generative AI use and misuse call for assessment reform in higher education

Growing misuse of, and unequal access to, AI tools requires universities to rethink how they evaluate learning

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The debate about the impact of generative artificial intelligence (GenAI) on higher education is polarized. Some portray GenAI as normalizing cheating at scale (*1*), whereas others argue that misconduct patterns have changed little (*2*). These competing narratives underscore the need for reliable data on where GenAI use is concentrated and where misuse is most likely. Existing studies of GenAI adoption and perceptions in higher education provide useful early signals but often rely on small samples and lack measures designed to capture sensitive behaviors such as cheating across fields (*3–5*). We addressed this gap with survey data from 95,513 students in a representative sample of 20 major public research-intensive universities in the United States and an indirect method for estimating GenAI-assisted cheating across disciplines. We found substantial heterogeneity in GenAI use and misuse across disciplines and student groups. These patterns call for discipline-specific assessment reform, not blanket bans or universal detection regimes.

GenAI is making some common forms of higher education assessment less reliable as evidence of student capability. Although these tools can support learning through personalized assistance and feedback, they can also function as cognitive shortcuts that allow students to outsource parts of the work that assessments are meant to evaluate (*6–8*). By making it easier to generate natural language, code, and audiovisual media, GenAI threatens the validity of commonly used assessments, which underpin the credibility of academic credentials and trust in higher education institutions (*9*).

Because the risks GenAI poses to assessment validity are likely to vary across disciplines and student subpopulations, identifying where use and misuse are concentrated is important for designing effective responses. We analyzed survey data collected between March and August 2024 [for analytic details, see supplementary materials (SM)]. Although our sample represents research universities—a relatively small subset of US higher-education institutions—these institutions enroll more than 2.6 million undergraduates and award more than half of all bachelor's degrees in science, technology, engineering, and mathematics (STEM) nationwide.

GENAI USE AND MISUSE

Our analysis reveals widespread GenAI use among students: Two-thirds reported using GenAI during the 2023–2024 academic year, and 37% reported using it regularly (i.e., monthly or more). Usage patterns differ considerably across disciplines (see the figure), with higher adoption in STEM fields, where computational skills and technology are instrumental to the curriculum. For instance, 62% of computer science students used GenAI regularly, compared with only 24% of students in the arts.

Notably, some social science disciplines also show high levels of GenAI adoption: Business students report a 51% use rate, and economics stands at 49%, even as related disciplines such as political science show lower rates (28%). This suggests that disciplines with analytically intensive coursework and data-driven assessments may be more conducive to GenAI integration, although these patterns may also reflect differences in the students who enroll in those fields.

In response to widespread GenAI use among students and emerging concerns about academic misconduct, many higher-education leaders have implemented restrictive policies, such as banning GenAI in assessments and enforcing disciplinary measures for unauthorized use. However, the effectiveness of these approaches is limited because GenAI-assisted cheating is inherently challenging to detect. Text-based detection methods are imperfect and likely to miss GenAI use when AI-assisted text is substantially edited, underscoring detection as an evolving cat-and-mouse problem rather than a settled technical solution (*10*). Accurate estimates of GenAI-assisted cheating are essential for designing and evaluating academic policies.

To estimate GenAI-assisted cheating, we used a list randomization experiment, an indirect-questioning technique designed to measure sensitive behaviors that respondents may be reluctant to disclose in direct questioning (*11*). This method allows respondents to answer sensitive questions anonymously, reducing social desirability bias. Students were randomly assigned to one of two groups in the survey. One group received three nonsensitive statements about using GenAI (e.g., "I have often explained to my classmates how to use AI tools like ChatGPT"), whereas the other received the same three statements plus a sensitive statement: "I have submitted AI-generated content as my own work in class, knowing it may not be allowed." Respondents reported only how many statements were true for them, not which ones. Comparing responses across groups allowed us to estimate, using maximum likelihood (*11*), the proportion of students engaging in GenAI-assisted cheating. Because some students may not recognize that their GenAI use constitutes a violation, this estimate is likely conservative.

We estimate that 9% of students who use GenAI tools have submitted AI-generated content despite knowing it may not be allowed (see the figure). This estimate is notably higher among daily GenAI users (26%) than among those who use it monthly (7%), suggesting that as the adoption of GenAI grows, we may see an upward trend in academic integrity violations. Although our estimates indicate that GenAI-assisted cheating may not be as widespread as anecdotal reports suggest (*1*), they underscore the considerable challenge it poses for maintaining academic standards.

GenAI-assisted cheating varies across disciplines. Overall, estimated rates are higher in non-STEM than in STEM fields. At the discipline level, for example, economics (17%) and journalism (16%) show relatively high rates, whereas biology is among the lowest, at 5%. Given that GenAI adoption in non-STEM disciplines is presently low, there is a concern that the absolute number of cheating cases could substantially increase as usage grows.

We also found large disparities in GenAI usage by gender, race or ethnicity, socioeconomic status, and disability status (see fig. S1). Only 33% of female students report regular GenAI use compared with 45% of male students (regression-adjusted odds ratio = 0.66). Similarly, usage is lower among underrepresented racial minorities (29%; see definition in SM) than among White and Asian students (39%; odds ratio = 0.71). Although socioeconomic status and disability status also show usage disparities, these gaps are less pronounced, and there are no significant differences by first-generation college status. These findings highlight concerns about equitable access to technological resources in higher education, as students from underrepresented backgrounds may have reduced access to or familiarity with GenAI.

GenAI usage disparities also vary across disciplines, with gender disparities most pronounced in health sciences and economics and racial-ethnic disparities most pronounced in the arts, humanities, and computer science. These variations suggest that differences in GenAI use are not uniform across fields and may reflect a combination of disciplinary contexts, enrollment composition, familiarity with technology, and other factors. This underscores the need for tailored approaches to equitable GenAI access and literacy across fields.

ASSESSMENT REFORM AS A POLICY RESPONSE

Our findings offer large-scale, discipline-specific evidence of GenAI use and cheating to better understand how students engage with this technology. Unlike previous small-sample surveys that did not account for disciplinary differences and generalized findings across fields, our survey captures substantial variation across disciplines. We found high adoption rates of GenAI in STEM fields but higher rates of GenAI-assisted cheating in non-STEM fields. Our list experiment also provides a robust estimate of the scale of GenAI-assisted cheating, showing that misconduct is concentrated among frequent GenAI users, who are three times as likely to cheat (26%) as less frequent users (7%). These findings suggest that GenAI-assisted misconduct is neither universal nor negligible. Although the estimated prevalence is lower than some anecdotal accounts suggest, the concentration of misuse among high-frequency users poses a substantial challenge for academic integrity policies as GenAI use continues to expand across fields.

The observed patterns matter not only because they indicate possible policy violations but also because they expose vulnerabilities in how universities infer student learning from assessment outcomes. As GenAI use grows, our findings suggest that some commonly used assessments may be becoming less dependable indicators of student capability, especially when they evaluate polished final outputs without sufficient evidence of process, judgment, and independent performance. This presents a direct challenge to assessment validity in disciplines where the capabilities being assessed can be partially outsourced to AI systems. Sweeping attempts to curb misuse can introduce new threats to validity when they rely on poorly justified restrictions, narrow assessment formats, or measures that impose unequal burdens across students.

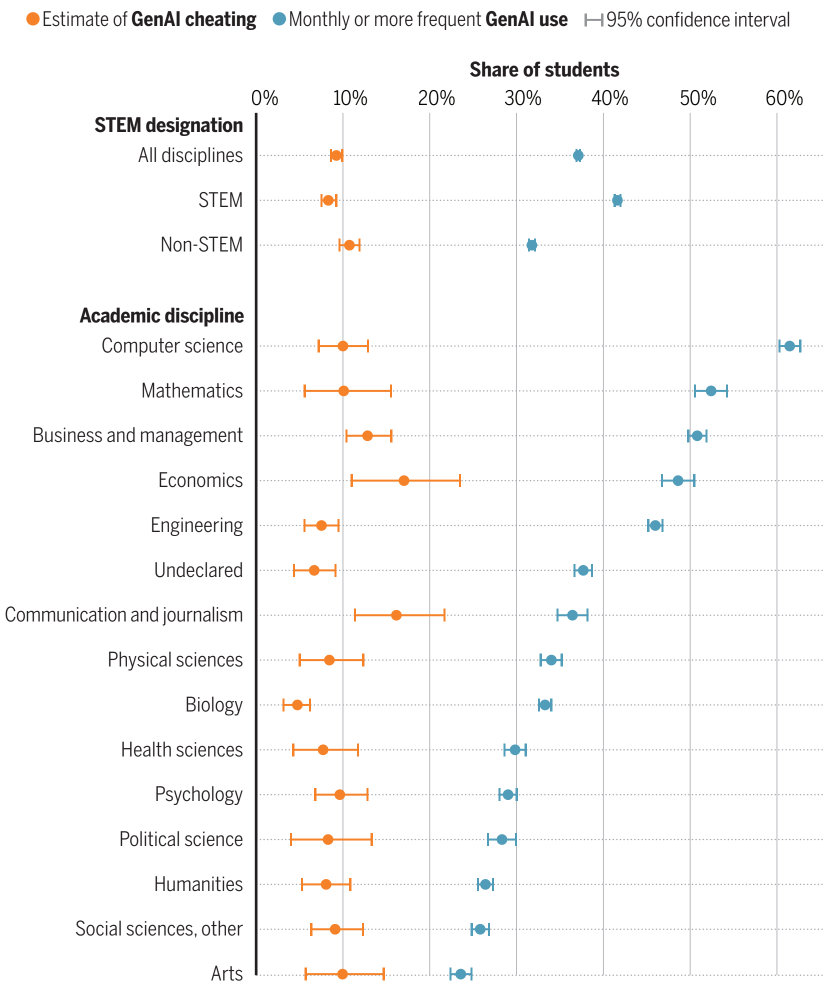
Assessment research therefore points not to bans and detection as the solution, but rather to assessment reform that strengthens the credibility of inferences about what students know and can do (*12*–*14*). Without such reforms, the credibility of assessments and trust in higher education credentials may erode further.

CHANGING HOW AND WHAT WE ASSESS

Central to these reforms is the revision of assessments in response to growing integrity risks associated with GenAI use. Our results suggest that reforms should be implemented at the discipline level, as each discipline combines distinct learning goals, assessment traditions, and ways in which GenAI can support or substitute for student work. This work may be best coordinated not by individual universities acting alone, but by disciplinary societies, accreditors, and other field-level organizations that can develop shared guidance and examples for local adaptation. Assessment

GenAI use and cheating by academic discipline

Blue dots show the share of students in each group who reported using generative artificial intelligence (GenAI) monthly or more frequently, based on the direct survey response (n = 95,513). Orange dots show maximum likelihood estimates of GenAI-assisted cheating derived from the list experiment among students who reported any GenAI use (n = 61,509).



STEM, science, technology, engineering, and mathematics.

GRAPHIC: V. PENNEY/*SCIENCE*

research increasingly emphasizes that there is no single "AI-proof" solution; instead, institutions must balance multiple goals: assuring learning, preserving authentic and meaningful assessment, making expectations enforceable, and preparing students for responsible AI use in professional contexts (*13*, *14*).

One response is to use more controlled assessment settings for selected learning outcomes, such as in-class, oral, practical, or test center–based assessments where independent performance can be observed more directly. These approaches may be especially useful when programs need stronger assurance that students can perform without external assistance. However, recent work cautions against treating a return to conventional exams as a general solution. Assessments that are ostensibly GenAI-proof can provide a partial short-term response, but they rarely capture the broader knowledge, judgment, and professional capabilities that many programs seek to develop (*13*). Controlled conditions should therefore be used selectively where they align with the competencies being assessed. Because controlled settings are limited in what they can measure, longer-term reform will also require clearer norms and redesigned assessments.

A second response is to clarify what constitutes acceptable and unacceptable AI use in specific assessment contexts. Institution-wide permissions or bans are insufficient because GenAI blurs the boundary between assistance and delegation across stages of work. Recent research shows that both students and instructors struggle to determine where the line should be drawn and that assessment policy must move beyond binary prohibition versus permission frameworks to consider enforceability, authentic learning, and the practical burden placed on teachers and students (*14*). Clearer guidance at the level of particular tasks and courses is therefore essential, especially where AI may be used for brainstorming, editing, coding, or feedback but not for replacing core disciplinary thinking.

A third response is to redesign assessments so that GenAI use is either structurally constrained by the task or purposefully incorporated into it. For example, requiring students to document their process, justify their choices, critique AI outputs, or demonstrate understanding through follow-up interaction shifts the focus away from the final artifact and toward the underlying reasoning. These design strategies also support the development of AI literacy, which is increasingly relevant to professional practice and expected from graduates. However, designing GenAI-integrated assignments is resource intensive and requires substantial faculty support, making implementation uneven across institutions. This reinforces the need for discipline-specific responses because differences in estimated misuse likely reflect variation in how disciplinary learning goals and assessment formats allow GenAI to substitute for the capabilities being assessed (*14*).

Given these complexities and resource constraints, universities will need to prioritize where reform efforts begin. Our prevalence estimates provide one useful input by identifying where existing exposure to GenAI use and misuse is greatest, but they should not be the sole criterion. Institutions should consider the consequences of incorrect judgments about student competence, the professional stakes attached to the competencies being assessed, and the feasibility of implementing valid alternatives in local contexts (*13*, *14*). Thus, fields with high rates of use and misuse may be practical starting points for reform, but lower-prevalence disciplines tied to especially high-stakes judgments may warrant equally urgent attention. Finally, recognizing that our data are drawn from US research universities, we expect that assessment reforms will require local adaptation across different national and institutional contexts.

ADDRESSING DISPARITIES AND FACULTY PREPAREDNESS

Closing sociodemographic gaps is crucial for the success of assessment reform because access to GenAI tools is increasingly essential in both academic and professional contexts. Our findings reveal that low-income, racially underrepresented, and female students are significantly less likely to use GenAI regularly than their peers. These disparities do not imply that some groups should use GenAI more, because higher use is not inherently better. Instead, they raise concerns about unequal opportunity to develop the literacy and critical oversight that are increasingly needed when GenAI shapes coursework, assessment, and professional practice. Universities should focus on ensuring that students are not differentially disadvantaged by providing institutionally supported tools and opportunities to learn when and how GenAI should be used responsibly, especially when reformed assessments assume that students can appropriately incorporate AI assistance. Equitable GenAI access and literacy can help prevent assessment reform from reproducing existing inequalities while also preparing students to engage meaningfully with the technologies that are shaping their future workplaces.

Faculty development is an important enabling condition for successful assessment reform and for bridging the gap between student adoption of GenAI and faculty readiness to adapt their teaching. Many instructors remain unprepared to address the challenges and opportunities presented by GenAI. Training should help educators understand the capabilities and limitations of these tools, design assessments that better capture student learning, communicate clearer expectations about acceptable use, and integrate GenAI into coursework where appropriate (*13*, *14*). These efforts will better equip faculty to implement assessment practices that reflect the realities of GenAI's impact on learning.

As GenAI becomes embedded in professional settings, universities must prepare students to use these tools responsibly while preserving credible ways of certifying human capability. This will require discipline-specific assessment reform, supported by more equitable access to GenAI and stronger faculty capacity to adapt teaching and assessment. Fair evaluation now depends not only on what is assessed but also on how evidence of learning is generated, interpreted, and bounded within particular disciplinary contexts. □

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SUPPLEMENTARY MATERIALS

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